Luke Y. Xia

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Education

Stanford University — 2026-2031
Ph.D. Physics — *GPA: N/A*
Advisors: TBD

University of California, Irvine — 2022 - 2026
B.S. Physics, B.S. Mathematics — *GPA: 3.97/4.00*
Advisor: James Bullock

Research Experience

Undergraduate Researcher, Theoretical Astrophysics — June 2026 - Present
Advisor: Kyle Kremer — *University of California, San Diego*

- Running massive Cluster Monte Carlo (CMC) simulations of globular clusters
- Co-leading a project on growing massive black holes in core-collapsed globular clusters
- Co-leading a project on building simulations to model observed Milky Way globular clusters

Undergraduate Researcher, Theoretical Astrophysics — September 2023 - Present
Advisor: James Bullock — *University of Southern California, University of California, Irvine*

- Led 3D shape evolution study of FIRE-simulated Milky Way-like galaxies, (1st-author); co-authored FIREbox morphology paper (3rd-author).
- Studying drivers of elongation within galaxies, drafting 1st-author paper.
- Mentored an undergraduate on morphology benchmarking across simulations vs. observations.
- Collaborating with Prof. Risa Wechsler (Stanford) on measuring subhalo spin distributions.

Undergraduate Researcher, Quantum Computing — September 2024 - Present
Advisors: Professor Sandy Irani, Professor Steven White — *University of California, Irvine*

- Built optimized localized QMA simulations in Julia (Monte Carlo + Lindbladian evolution).
- Implemented adaptive Krylov exponentiation for Liouvillian time evolution; computed spectral gaps (Arnoldi/Lanczos) for convergence bounds.

TREND NSF REU Intern, Computational Plasma Physics — May 2023 - August 2023
Advisor: Dr. Marc Swisdak — *University of Maryland, College Park*

- Ran `p3d` PIC reconnection simulations; developed Klein-bottle processor mapping cutting runtime 50% while preserving periodicity/reconnection rates.
- Developed novel boundary conditions in Fortran; published 1st-author paper in *Physics of Plasmas*.

Undergraduate Researcher, Observational Astroparticle Physics — December 2022 - March 2024
Advisor: Professor Simona Murgia — *University of California, Irvine*

- Analyzed Fermi-LAT data on the galactic center excess and dwarf spheroidals for SIDM constraints; processed 15 years of data with Python/FermiPy and FITS workflows.

First/Lead-Author Publications

1. “Magnetic Reconnection on a Klein bottle.”
   Luke Xia, M. Swisdak
   Physics of Plasmas, 31(8), 083902. (2024) https://doi.org/10.1063/5.0222454
2. “The Shape of FIREbox Galaxies and a Potential Tension with Low-mass Disks.”
   C. Klein, J. D. Wang, **Luke Xia**, J. S. Bullock, et al.
   The Astrophysical Journal, 998(1), 125. (2026) https://doi.org/10.3847/1538-4357/ae31f4
3. “Pickles on FIRE: The 3D Shape Evolution of Simulated Milky-Way Mass Galaxies.”
   Luke Xia, C. Klein, J. D. Wang, J. Bullock, et al.
   Monthly Notices of the Royal Astronomical Society. (2026) https://doi.org/10.1093/mnras/stag964
4. “On the Physics of Galaxy Elongation”(in prep)
   Luke Y. Xia, James Bullock, et al.

Contributed Publications

1. "From protogalaxy through thick and thin: Why did the Milky Way evolve in three kinematic phases?" (submitted to MNRAS)
   Olti Myrtaj, James Bullock, et al. (including **Luke Y. Xia**).
2. "Shape Distributions of Simulated Galaxies from High Redshift to the Present" (in prep)
   Jenny Wang, **Luke Y. Xia**, James Bullock, et al.

Presentations

12. Poster, AAS 248, Pasadena, CA	06/2026
11. Poster, UCI UROP Symposium, Irvine, CA	05/2026
10. Invited Talk, USC Cosmolab Seminar, Los Angeles, CA	04/2026
9. Poster, AAS 247, Phoenix, AZ	01/2026
8. Invited Talk, Gulf Coast Undergraduate Research Symposium, Houston, TX	09/2025
7. Talk, APS Global Summit, Anaheim, CA	03/2025
6. Talk, AAS 245, National Harbor, MD	01/2025
5. Talk, GALFRESCA, Pasadena, CA	09/2024
4. Poster, UCI UROP Symposium, Irvine, CA	05/2024
3. Poster, APS 2024 March Meeting, Minneapolis, MN	03/2024
2. Poster, NSBP Annual Meeting, Knoxville, TN	10/2023
1. Poster, GRAD-MAP/TREND 2023 Scholars Conference, College Park, MD	07/2023

Honors and Awards

- Stanford KIPAC Fellowship — 2029-2031
- Stanford Graduate Fellowship — 2026-2029
- National Science Foundation Graduate Research Fellowship — 2026-2029
- Barry Goldwater Scholarship (press) — 2025
- LLNL/AIP Leadership Scholarship (press) — 2025
- Rose Hills Scholarship — 2025
- Distinguished Anteater Scholarship — 2025
- Outstanding Non-Graduating Undergraduate Research Award — 2025
- Bullock Group Research Grant — 2024-2026
- UCI Undergraduate Research Opportunities Program: Summer Undergraduate Research Program (2x), Travel Grant (2x), Research Experience Fellowship (1x) — 2023-2026
- TREND NSF REU Research Grant & Travel Award — 2023

K-12 Outreach

As the President of the Society of Physics Students, I created an outreach team that visited 20+ K-12 schools to present physics demonstrations and what it's like to study STEM in college. Listed are some of the trips that I led and attended.

- Burton Tech High School, Compton, CA — 05/2026
  Spoke to 5 classes on college life, applications, STEM/physics pathways, with physics demonstrations
- Beckman High School, Tustin, CA — 05/2026
  Spoke to 4 classes on college life, applications, STEM/physics pathways, with physics demonstrations
- Towngate Elementary School, Moreno Valley, CA — 05/2026
  Presented physics demonstrations to 5 classes
- OCASA Charter School, Laguna Niguel, CA — 04/2026
  Presented demonstrations at Science Fair
- Hillview Continuing Education High School, Tustin, CA — 03/2026
  Presented demonstrations to 3 classes
- Cypress High School, Cypress, CA — 01/2026
  Spoke to 4 classes on college life, applications, STEM/physics pathways, with physics demonstrations

- Eastbluff Elementary School, Newport Beach, CA — 09/2025
  Presented demonstrations at Science Fair
- Summit High School, Fontana, CA — 05/2025
  Spoke to 4 classes on college life, applications and STEM/physics pathways
- Laguna Beach High School, Laguna Beach, CA — 04/2025
  Presented physics demonstrations to 3 classes

Skills

Programming Languages (In order of proficiency): Python, Julia, Fortran, IDL, Matlab, JavaScript, C++, C
Operating Systems/Computing Clusters: Windows, macOS, Linux, High Performance Computing (NERSC Perlmutter, Greenplanet), Slurm Workload Manager
Software Packages/Libraries: `p3d`, `FLASH`, `GIZMO`, Mathematica, R Studio, SAOImage DS9, LaTeX

Selected Coursework

327/180 units completed, including 13 graduate courses (†).

Graduate Physics: Classical Mechanics; Mathematical Physics; Quantum Mechanics I; Quantum Field Theory; Plasma Physics II; Galactic Astrophysics; Radiative Processes; Electromagnetic Theory I/II; Statistical Mechanics; Tensor Networks & Quantum Simulation; Machine Learning & Statistics.
Core Physics: Classical Mechanics I/II; Electromagnetic Theory I/II; Quantum Mechanics; Statistical Mechanics; Mathematical Physics I/II; Introduction to Cosmology; Astrophysics of Galaxies; Observational Astrophysics; High-Energy Astrophysics.
Mathematics: Abstract Algebra I/II; Advanced Linear Algebra I/II; Analysis I/II; Complex Analysis; Number Theory I/II; Algebraic Coding Theory.

Leadership and Extracurriculars

Society of Physics Students at UCI — September 2022 - Present
- Webmaster (2023–2024), President (2024–2025), Outreach Chair (2025–2026) for the UCI SPS chapter.
- Scaled operations by expanding the executive board 4→18 and launching an 8-person outreach team.
- Registered 116 UCI students as national SPS members and increased average meeting attendance 15 to 40.+
- Planned and led 31 weekly events and 26 board meetings in collaboration with the Physics Department and campus Physical Sciences organizations.
- Founded major programs: Big Dipper Little Dipper undergrad–grad mentorship (36 participants in year 1), undergraduate research talk series (12 talks), and a cross-club Friendsgiving across 8+ organizations.
- Initiated inter-chapter collaborations with Cal Poly Pomona and UC San Diego, including a nuclear reactor tour and a 15-person graduate applications panel.

National Society of Black Physicists — September 2023 - Present
- Founding member and Secretary of California's first NSBP chapter; coordinated UCI participation in 2023 and 2024 annual conferences.
- Led joint outreach with SPS to local high schools and community colleges, reaching 27 classes across 11 schools.

Physics Resonance Mentor — September 2023 - Present
- Mentored incoming freshman and transfer students (Fall 2023–2025) on research onboarding, cold emails, and professor matching.

Other Activities
- Physics Advising Subcommittee (2023–2024); Quantum Computing @ UCI; Undergraduate Mathematics Committee.
- UCI Club Badminton Team (2023–2024); CTSA Symphony Orchestra (violin, 2022–2023).
- Boy Scouts of America: Eagle Scout; led virtual music lessons project (300+ hours) during COVID-19.
- Math Teaching Assistant, Art of Problem Solving (2020): taught competitive math camps; built a 20-player cryptography game platform (HTML/CSS/JS).